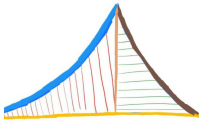
Math for EE



G. V. V. Sharma

About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. CBSE CLASS 12 MATHEMATICS (65/7/1)

A. Formulae

1.1.1 Domain of inverse functions. $\cos^{-1} x$ and $\sin^{-1} x$ are defined only for $-1 \le x \le 1$, while $\sin x$ and $\cos x$ are defined for all real x. The domain of a sum is the intersection of the domains.

1.1.2 Counting. If each of m independent positions can take c values, the total number of arrangements is

$$N = c^m \tag{1.1.2.1}$$

1.1.3 Symmetric matrix. A square matrix is symmetric if

$$\mathbf{A}^\top = \mathbf{A} \tag{1.1.3.1}$$

Every diagonal matrix is symmetric.

1.1.4 Determinant properties. For $n \times n$ matrices $\mathbf{A}$, $\mathbf{B}$ and a scalar k,

$$|k\mathbf{A}| = k^n |\mathbf{A}|, \qquad |\mathbf{AB}| = |\mathbf{A}|\,|\mathbf{B}| \tag{1.1.4.1}$$

1.1.5 Adjugate. With $\mathbf{A}\,\mathrm{adj}\,\mathbf{A} = |\mathbf{A}|\,\mathbf{I}$,

$$|\mathrm{adj}\,\mathbf{A}| = |\mathbf{A}|^{n-1} \tag{1.1.5.1}$$

Solution: Taking determinants of $\mathbf{A}\,\mathrm{adj}\,\mathbf{A} = |\mathbf{A}|\,\mathbf{I}$ gives $|\mathbf{A}|\,|\mathrm{adj}\,\mathbf{A}| = |\mathbf{A}|^n$.

1.1.6 Characteristic polynomial and Cayley–Hamilton theorem. For a 2×2 matrix,

$$\lambda^2 - \mathrm{tr}\,(\mathbf{A})\,\lambda + |\mathbf{A}| = 0, \qquad \mathbf{A}^2 - \mathrm{tr}\,(\mathbf{A})\,\mathbf{A} + |\mathbf{A}|\,\mathbf{I} = \mathbf{0} \tag{1.1.6.1}$$

The eigenvalues sum to the trace and multiply to the determinant.

1.1.7 Cramer's rule. For $\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} e \\ f \end{pmatrix}$ with $\Delta = ad - bc \neq 0$,

$$x = \frac{ed - bf}{\Delta}, \quad y = \frac{af - ec}{\Delta} \tag{1.1.7.1}$$

1.1.8 Monotonicity. If $f'(x) > 0$ for all x in an interval, f is strictly increasing there; if $f'(x) < 0$, it is strictly decreasing.

1.1.9 Logarithm and exponential. For $x > 0$,

$$e^{k \log x} = x^k \tag{1.1.9.1}$$

1.1.10 Integral properties.

$$\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx, \qquad \int_{-a}^{a} f(x)\,dx = \begin{cases} 2\int_0^a f(x)\,dx & f \text{ even} \\ 0 & f \text{ odd} \end{cases} \tag{1.1.10.1}$$

$$\int_0^{2a} f(x)\,dx = \int_0^a f(x)\,dx + \int_0^a f(a+x)\,dx \tag{1.1.10.2}$$

1.1.11 Midpoint Riemann sum. With n subintervals of width $\Delta x = \frac{b-a}{n}$ and midpoints $x_i^* = a + \left(i - \frac{1}{2}\right)\Delta x$,

$$\int_a^b f(x)\, dx \approx \sum_{i=1}^{n} f(x_i^*)\, \Delta x \tag{1.1.11.1}$$

The sum converges to the integral as $n \to \infty$.

1.1.12 Standard integrals,

$$\int \frac{du}{\sqrt{1-u^2}} = \sin^{-1} u + C, \qquad \int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + C \tag{1.1.12.1}$$

1.1.13 Derivatives. Product rule, chain rule and the exponential rule,

$$\frac{d}{dx}(uv) = u'v + uv', \quad \frac{d}{dx}a^x = a^x \ln a, \quad \frac{d}{dx}\log(\sin u) = \cot u \frac{du}{dx} \tag{1.1.13.1}$$

1.1.14 Symmetric integral. If $g(\pi - x) = g(x)$,

$$\int_0^{\pi} x\, g(x)\, dx = \frac{\pi}{2}\int_0^{\pi} g(x)\, dx \tag{1.1.14.1}$$

Solution: Substituting $x \to \pi - x$ gives $I = \int_0^{\pi}(\pi - x)\, g(x)\, dx$. Adding this to the original, $2I = \pi \int_0^{\pi} g(x)\, dx$.

1.1.15 Laplace transform. For a first-order equation $y' + ay = f(x)$,

$$\mathcal{L}\{y'\} = sY(s) - y(0), \quad \mathcal{L}\{\cos x\} = \frac{s}{s^2+1}, \quad \mathcal{L}\left\{e^{-ax}\right\} = \frac{1}{s+a} \tag{1.1.15.1}$$

$$\mathcal{L}\{1\} = \frac{1}{s}, \quad \mathcal{L}\{t\} = \frac{1}{s^2} \tag{1.1.15.2}$$

The order of a differential equation is the highest derivative present; the degree is its power.

1.1.16 Orthogonality and scalar product with a unit vector,

$$|\mathbf{a}+\mathbf{b}| = |\mathbf{a}-\mathbf{b}| \iff \mathbf{a}^{\top}\mathbf{b} = 0, \qquad \text{scalar product of } \mathbf{a} \text{ with unit vector along } \mathbf{v} = \frac{\mathbf{a}^{\top}}{\|\mathbf{v}} \tag{1.1.16.1}$$

1.1.17 Distance between parallel lines $\mathbf{x} = \mathbf{P}_1 + \lambda\mathbf{d}$ and $\mathbf{x} = \mathbf{P}_2 + \mu\mathbf{d}$,

$$d = \frac{\|(\mathbf{P}_2 - \mathbf{P}_1) \times \mathbf{d}\|}{\|\mathbf{d}\|} \tag{1.1.17.1}$$

1.1.18 Points on a line in space. A line $\mathbf{x} = \mathbf{A} + t\mathbf{d}$ is written with one parameter t, and the distance of a point on it from $\mathbf{P}$ is $\|\mathbf{A} + t\mathbf{d} - \mathbf{P}\|$. Used in Q.35(A).

1.1.19 Foot of the perpendicular and image. For a point $\mathbf{P}$ and the line $\mathbf{A} + t\mathbf{d}$,

$$t_0 = \frac{(\mathbf{P} - \mathbf{A})^{\top}\mathbf{d}}{\mathbf{d}^{\top}\mathbf{d}}, \quad \mathbf{H} = \mathbf{A} + t_0\mathbf{d}, \quad \mathbf{P}' = 2\mathbf{H} - \mathbf{P} \tag{1.1.19.1}$$

1.1.20 Corner point theorem. The maximum or minimum of a linear objective over a feasible region occurs at a corner point. For an unbounded region, a minimum exists only if

the objective is bounded below there.

1.1.21 Conditional probability,

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}, \qquad \Pr(A' \cap B) = \Pr(B) - \Pr(A \cap B) \tag{1.1.21.1}$$

and $\Pr(A \cap B) \leq \min\{\Pr(A), \Pr(B)\}$.

1.1.22 Probability mapping (pmf). For an event with indicator $m(s) \in \{0, 1\}$ over outcomes with probabilities p_s,

$$\Pr(E) = \sum_s m(s)\, p_s, \qquad E[X] = \sum_x x \Pr(X = x) \tag{1.1.22.1}$$

For independent trials the joint pmf is the product of the individual pmfs.

1.1.23 Total probability and Bayes' theorem. For mutually exclusive, exhaustive events $A_1, \ldots, A_n$,

$$\Pr(D) = \sum_i \Pr(D \mid A_i) \Pr(A_i), \qquad \Pr(A_1 \mid D) = \frac{\Pr(D \mid A_1) \Pr(A_1)}{\Pr(D)} \tag{1.1.23.1}$$

B. Problems

1.2.1 Find the domain of the function $f(x) = \cos^{-1} x + \sin x$. (CBSE Class 12, 65/7/1, Q.2)

Solution: $\cos^{-1} x$ is defined only for $-1 \leq x \leq 1$, and $\sin x$ is defined for every real x. The domain of the sum is the intersection of the two domains:

$$\text{Domain}(f) = [-1, 1] \quad \text{(Option C)} \tag{1.2.1.1}$$

The program scans x over $[-3, 3]$ and treats the points where `acos(x)+sin(x)` is not a number as outside the domain. The following code verifies the solution

codes/2.c

1.2.2 The number of all possible matrices of order 3×3 with each entry 2 or 3 is (CBSE Class 12, 65/7/1, Q.3)

Solution: Each of the 9 entries can independently be 2 or 3. By the multiplication principle,

$$N = 2^9 = 512 \quad \text{(Option B)} \tag{1.2.2.1}$$

The following code verifies the solution

codes/3.c

1.2.3 The matrix $\mathbf{A} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}$ is a (CBSE Class 12, 65/7/1, Q.4)

a) scalar matrix b) identity matrix c) null matrix d) symmetric matrix

Solution: (theory only) A matrix is symmetric if $\mathbf{A}^\top = \mathbf{A}$. For a diagonal matrix the transpose leaves every entry unchanged,

$$\mathbf{A}^\top = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix} = \mathbf{A} \tag{1.2.3.1}$$

The diagonal entries are not all equal, so $\mathbf{A}$ is not a scalar matrix; it is neither the identity nor the null matrix. Hence $\mathbf{A}$ is a symmetric matrix (Option D).

1.2.4 If $\mathbf{A}$ and $\mathbf{B}$ are square matrices of order 3 with $|\mathbf{A}| = 3$ and $|\mathbf{B}| = 5$, then $|2\mathbf{AB}|$ is (CBSE Class 12, 65/7/1, Q.5)

Solution: For an $n \times n$ matrix, $|k\mathbf{A}| = k^n |\mathbf{A}|$ and $|\mathbf{AB}| = |\mathbf{A}|\,|\mathbf{B}|$. With $n = 3$,

$$|2\mathbf{AB}| = 2^3 |\mathbf{A}|\,|\mathbf{B}| = 8\,(3)\,(5) = 120 \quad \text{(Option B)} \tag{1.2.4.1}$$

Standard C has no built-in determinant routine, so the program defines `det3()` using cofactor expansion and applies it to the product matrix $2\mathbf{AB}$. The following code verifies the solution

codes/5.c

1.2.5 If $\mathbf{A}$ is a square matrix of order 3 with $|\mathbf{A}| = 5$, then $|\text{adj}\,\mathbf{A}|$ is (CBSE Class 12, 65/7/1, Q.6)

Solution: For an $n \times n$ matrix, $\mathbf{A}\,\text{adj}\,\mathbf{A} = |\mathbf{A}|\,\mathbf{I}$. Taking determinants,

$$|\mathbf{A}|\,|\text{adj}\,\mathbf{A}| = |\mathbf{A}|^n \implies |\text{adj}\,\mathbf{A}| = |\mathbf{A}|^{n-1} \tag{1.2.5.1}$$

For $n = 3$,

$$|\text{adj}\,\mathbf{A}| = 5^2 = 25 \quad \text{(Option C)} \tag{1.2.5.2}$$

The program builds $\text{adj}\,\mathbf{A}$ from cofactors and applies the same `det3()` function. The following code verifies the solution

codes/6.c

1.2.6 The function $f(x) = 2x - \sin x$, $x \in \mathbb{R}$, is (CBSE Class 12, 65/7/1, Q.11)

Solution: Differentiating,

$$f'(x) = 2 - \cos x \tag{1.2.6.1}$$

Since $-1 \le \cos x \le 1$,

$$1 \le f'(x) \le 3 \implies f'(x) > 0 \quad \forall x \in \mathbb{R} \tag{1.2.6.2}$$

A differentiable function with a strictly positive derivative is strictly increasing, so f is strictly increasing on $\mathbb{R}$ (Option B). The program classifies each scanned point with `if-else` on the sign of $f'(x)$. The following code verifies the solution

codes/11.c

The following code generates Fig. 1.2.6.1.

codes/11plot.py

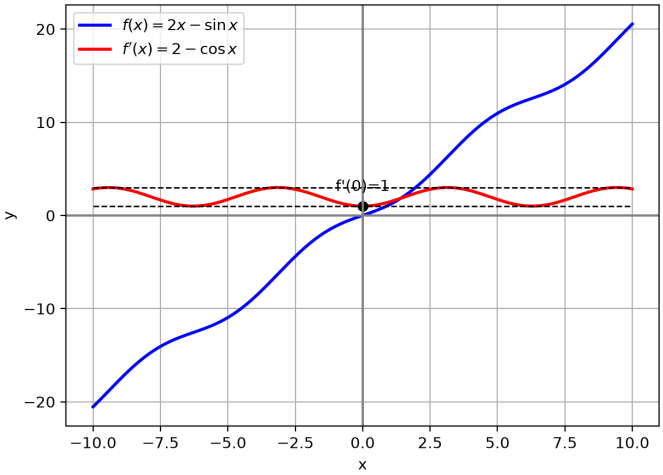


Fig. 1.2.6.1: $f(x)$ and $f'(x)$ with $1 \leq f'(x) \leq 3$

1.2.7 Evaluate $\int \frac{e^{9\log x} - e^{8\log x}}{e^{6\log x} - e^{5\log x}}\,dx$. (CBSE Class 12, 65/7/1, Q.12)

Solution: For $x > 0$, $e^{k\log x} = x^k$. Hence, for $x \neq 1$,

$$\frac{x^9 - x^8}{x^6 - x^5} = \frac{x^8(x-1)}{x^5(x-1)} = x^3 \tag{1.2.7.1}$$

$$\int x^3\,dx = \frac{x^4}{4} + C \quad \text{(Option C)} \tag{1.2.7.2}$$

The midpoint Riemann sum $\sum_{i=1}^{n} f(x_i^*)\,\Delta x$ of the original integrand over $[2,4]$ is compared with $\frac{x^4}{4}\Big|_2^4 = 60$. The following code verifies the solution

codes/12.py

1.2.8 Which of the following statements is correct? (CBSE Class 12, 65/7/1, Q.13)

(A) $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$

(B) $\int_{-a}^{a} f(x)\,dx = 0$ if f is even

(C) $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$ if f is odd

(D) $\int_{0}^{2a} f(x)\,dx = \int_{0}^{a} f(x)\,dx - \int_{0}^{a} f(2a+x)\,dx$

Solution: **(A)** Put $u = a+b-x$, $du = -dx$. When $x = a$, $u = b$ and when $x = b$, $u = a$, so

$$\int_{a}^{b} f(a+b-x)\,dx = \int_{b}^{a} f(u)\,(-du) = \int_{a}^{b} f(u)\,du \tag{1.2.8.1}$$

(A) is true.

(B) For an even function $\int_{-a}^{a} f = 2\int_{0}^{a} f$. With $f(x) = x^2$, $\int_{-1}^{1} x^2\,dx = \frac{2}{3} \neq 0$. False.

(C) For an odd function $\int_{-a}^{a} f = 0$. With $f(x) = x$, the left side is 0 and the right side is 1. False.

(D) The correct splitting is $\int_{0}^{2a} f = \int_{0}^{a} f + \int_{0}^{a} f(a+x)\,dx$. With $f(x) = x$, $a = 1$, the left side is 2 and the right side is -2. False.

Hence the correct statement is (A). The program checks each option with SymPy. The following code verifies the solution

codes/13.py

1.2.9 Evaluate $\int \frac{e^x}{\sqrt{4-e^{2x}}}\,dx$. (CBSE Class 12, 65/7/1, Q.14)

Solution: Let $u = \frac{e^x}{2}$, so $du = \frac{e^x}{2}\,dx$ and $4 - e^{2x} = 4\left(1-u^2\right)$. Then

$$\int \frac{e^x\,dx}{\sqrt{4-e^{2x}}} = \int \frac{2\,du}{2\sqrt{1-u^2}} = \sin^{-1} u + C = \sin^{-1}\left(\frac{e^x}{2}\right) + C \quad \text{(Option D)} \tag{1.2.9.1}$$

valid for $e^{2x} < 4$, i.e. $x < \ln 2$. The program compares a midpoint Riemann sum over $[-1, 0.5]$ with $F(b) - F(a)$. The following code verifies the solution

codes/14.py

1.2.10 If $\mathbf{a}$ and $\mathbf{b}$ are two vectors such that $|\mathbf{a}+\mathbf{b}| = |\mathbf{a}-\mathbf{b}|$, then $\mathbf{a}$ and $\mathbf{b}$ are (CBSE Class 12, 65/7/1, Q.16)

Solution: (theory only) Squaring both sides and using $|\mathbf{x}|^2 = \mathbf{x}^\top\mathbf{x}$,

$$|\mathbf{a}|^2 + 2\mathbf{a}^\top\mathbf{b} + |\mathbf{b}|^2 = |\mathbf{a}|^2 - 2\mathbf{a}^\top\mathbf{b} + |\mathbf{b}|^2 \tag{1.2.10.1}$$

$$\implies 4\mathbf{a}^\top\mathbf{b} = 0 \tag{1.2.10.2}$$

Hence $\mathbf{a}$ and $\mathbf{b}$ are orthogonal (Option A); the zero vector is the trivial case.

1.2.11 If $\Pr(A) = \frac{1}{7}$, $\Pr(B) = \frac{5}{7}$ and $\Pr(A \cap B) = \frac{4}{7}$, find $\Pr(A' \mid B)$. (CBSE Class 12, 65/7/1, Q.17)

Solution: By the definition of conditional probability,

$$\Pr(A' \mid B) = \frac{\Pr(A' \cap B)}{\Pr(B)} = \frac{\Pr(B) - \Pr(A \cap B)}{\Pr(B)} \tag{1.2.11.1}$$

This needs only $\Pr(B)$ and $\Pr(A \cap B)$:

$$\Pr(A' \mid B) = \frac{\frac{5}{7} - \frac{4}{7}}{\frac{5}{7}} = \frac{1}{5} \tag{1.2.11.2}$$

The values as printed are not consistent with each other, because an intersection cannot be more probable than A alone ($\frac{4}{7} > \frac{1}{7}$); the program prints a warning for this and then evaluates the formula above. The following code verifies the solution

codes/17.c

1.2.12 A coin is tossed and a card is selected at random from a well-shuffled pack of 52 cards. The probability of getting a head and a face card is (CBSE Class 12, 65/7/1, Q.18)

Solution: Define the indicator maps

$$X = \begin{cases} 1 & \text{head} \\ 0 & \text{tail} \end{cases}, \qquad Y = \begin{cases} 1 & \text{face card} \\ 0 & \text{otherwise} \end{cases} \tag{1.2.12.1}$$

There are 12 face cards, so $\Pr(X = 1) = \frac{1}{2}$ and $\Pr(Y = 1) = \frac{12}{52} = \frac{3}{13}$. The two trials are independent, so the joint pmf is the product,

$$\Pr(X = 1, Y = 1) = \frac{1}{2} \cdot \frac{3}{13} = \frac{3}{26} \quad \text{(Option B)} \tag{1.2.12.2}$$

The following code verifies the solution

codes/18.c

1.2.13 If $\mathbf{A} = \begin{pmatrix} 2 & 3 \\ -1 & 2 \end{pmatrix}$, show that $\mathbf{A}^2 - 4\mathbf{A} + 7\mathbf{I} = \mathbf{0}$. (CBSE Class 12, 65/7/1, Q.22)

Solution: The trace and determinant are

$$\text{tr}(\mathbf{A}) = 2 + 2 = 4, \qquad |\mathbf{A}| = 2(2) - 3(-1) = 7 \tag{1.2.13.1}$$

The characteristic polynomial of a 2×2 matrix is $\lambda^2 - \text{tr}(\mathbf{A})\,\lambda + |\mathbf{A}|$, so

$$\lambda^2 - 4\lambda + 7 = 0 \implies \lambda = \frac{4 \pm \sqrt{16 - 28}}{2} = 2 \pm i\sqrt{3} \tag{1.2.13.2}$$

The eigenvalues sum to 4 (the trace) and multiply to 7 (the determinant). By the Cayley–Hamilton theorem, $\mathbf{A}$ satisfies its own characteristic equation:

$$\mathbf{A}^2 - 4\mathbf{A} + 7\mathbf{I} = \mathbf{0} \tag{1.2.13.3}$$

Direct verification: $\mathbf{A}^2 = \begin{pmatrix} 1 & 12 \\ -4 & 1 \end{pmatrix}$, $4\mathbf{A} = \begin{pmatrix} 8 & 12 \\ -4 & 8 \end{pmatrix}$, $7\mathbf{I} = \begin{pmatrix} 7 & 0 \\ 0 & 7 \end{pmatrix}$, so

$$\mathbf{A}^2 - 4\mathbf{A} + 7\mathbf{I} = \begin{pmatrix} 1-8+7 & 12-12 \\ -4+4 & 1-8+7 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \tag{1.2.13.4}$$

The following code verifies the solution

codes/22.c

1.2.14 Differentiate $y = \dfrac{5^x}{x^5}$ with respect to x. (CBSE Class 12, 65/7/1, Q.23(A))

Solution: Write $y = 5^x x^{-5}$ and use the product rule with $\frac{d}{dx}a^x = a^x \ln a$:

$$\frac{dy}{dx} = 5^x \ln 5 \cdot x^{-5} + 5^x \left(-5x^{-6}\right) = \frac{5^x}{x^5}\left(\ln 5 - \frac{5}{x}\right) \tag{1.2.14.1}$$

The following code verifies the solution

codes/23.py

1.2.15 Maximise $Z = 5x + 4y$ subject to $3x + y \le 6$, $x \le 1$, $x \ge 0$, $y \ge 0$. (CBSE Class 12, 65/7/1, Q.24)

Solution: The boundary lines are $3x + y = 6$, $x = 1$, $x = 0$, $y = 0$. The feasible region is bounded by the corner points

$$(0,0)\,,\ (1,0)\,,\ (1,3)\,,\ (0,6) \tag{1.2.15.1}$$

Corner point (x, y)	$Z = 5x + 4y$
$(0, 0)$	0
$(1, 0)$	5
$(1, 3)$	17
$(0, 6)$	24

TABLE 1.2.15: Value of Z at the corner points

The maximum value is

$$Z_{\max} = 24 \text{ at } (x, y) = (0, 6) \tag{1.2.15.2}$$

The following code verifies the solution

codes/24.c

The following code generates Fig. 1.2.15.1.

codes/24plot.py

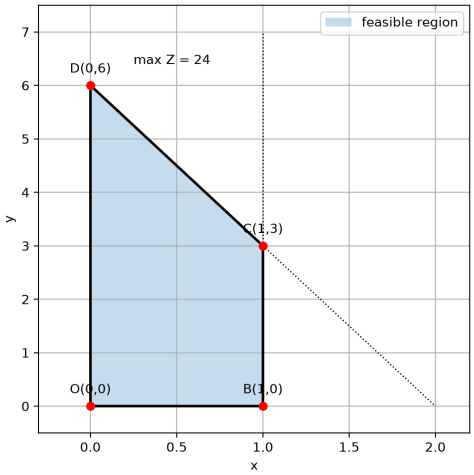


Fig. 1.2.15.1: Feasible region and corner points

1.2.16 A box contains 10 blocks: two marked 0, three marked 1, four marked 2 and one marked 3. A block is drawn at random. Find the probability distribution of the number X on the block and its mean. (CBSE Class 12, 65/7/1, Q.25(A))

Solution: Each block is equally likely, so the pmf is $\Pr(X=x) = \frac{\text{number of blocks marked } x}{10}$:

x	0	1	2	3
$\Pr(X=x)$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{4}{10}$	$\frac{1}{10}$

TABLE 1.2.16: Probability mass function of X

The probabilities sum to 1. The mean is

$$E[X] = \sum_x x \Pr(X=x) = 0\left(\frac{2}{10}\right) + 1\left(\frac{3}{10}\right) + 2\left(\frac{4}{10}\right) + 3\left(\frac{1}{10}\right) = \frac{14}{10} = 1.4 \tag{1.2.16.1}$$

The following code verifies the solution

codes/25.c

1.2.17 Two ants move along the lines $2x + 5y - 1 = 0$ and $3x + 2y - 7 = 0$. Using the matrix method, find the point where their paths meet. (CBSE Class 12, 65/7/1, Q.27(A))

Solution: In matrix form,

$$\begin{pmatrix} 2 & 5 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 7 \end{pmatrix} \tag{1.2.17.1}$$

The determinant is $2(2)-5(3)=-11\neq 0$, so the solution is unique. By Cramer's rule,

$$x=\frac{1(2)-5(7)}{-11}=3,\qquad y=\frac{2(7)-1(3)}{-11}=-1 \tag{1.2.17.2}$$

The paths meet at $(3,-1)$. The following code verifies the solution

codes/27.c

1.2.18 Find $\frac{dy}{dx}$ if $y=\sqrt{\log\left(\sin\left(\frac{x^3}{3}-1\right)\right)}$. (CBSE Class 12, 65/7/1, Q.28)

Solution: Let $u=\frac{x^3}{3}-1$, so $\frac{du}{dx}=x^2$. By the chain rule,

$$\frac{dy}{dx}=\frac{1}{2\sqrt{\log(\sin u)}}\cdot\frac{d}{dx}\log(\sin u)=\frac{1}{2\sqrt{\log(\sin u)}}\cdot\frac{\cos u}{\sin u}\cdot x^2 \tag{1.2.18.1}$$

$$=\frac{x^2\cot\left(\frac{x^3}{3}-1\right)}{2\sqrt{\log\left(\sin\left(\frac{x^3}{3}-1\right)\right)}} \tag{1.2.18.2}$$

The following code verifies the solution

codes/28.py

1.2.19 Minimise $Z=18x+10y$ subject to $4x+y\geq 20$, $2x+3y\geq 30$, $x\geq 0$, $y\geq 0$. (CBSE Class 12, 65/7/1, Q.30)

Solution: The lines $4x+y=20$ and $2x+3y=30$ meet where $y=20-4x$ and $2x+3(20-4x)=30$, so $x=3$, $y=8$. The feasible region is unbounded, with corner points $(0,20)$, $(3,8)$ and $(15,0)$:

$$Z(0,20)=200,\qquad Z(3,8)=54+80=134,\qquad Z(15,0)=270 \tag{1.2.19.1}$$

The minimum value is

$$Z_{\min}=134 \text{ at } (3,8) \tag{1.2.19.2}$$

The following code verifies the solution

codes/30.c

The following code generates Fig. 1.2.19.1.

codes/30plot.py

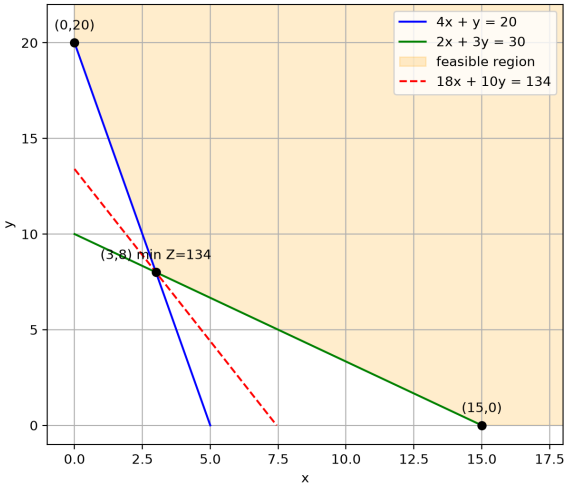


Fig. 1.2.19.1: Unbounded feasible region and the line $Z = 134$

1.2.20 Let $\mathbf{a} = \hat{i} - \hat{j} + 2\hat{k}$, $\mathbf{b} = 2\hat{i} - 4\hat{j} + 5\hat{k}$ and $\mathbf{c} = \lambda\hat{i} - 2\hat{j} - 3\hat{k}$. Find λ if the scalar product of $\mathbf{a}$ with the unit vector along $\mathbf{b} + \mathbf{c}$ is 1. (CBSE Class 12, 65/7/1, Q.31(A))

Solution:

$$\mathbf{b} + \mathbf{c} = \begin{pmatrix} \lambda + 2 \\ -6 \\ 2 \end{pmatrix}, \qquad |\mathbf{b} + \mathbf{c}| = \sqrt{(\lambda + 2)^2 + 40} \tag{1.2.20.1}$$

The scalar product with $\mathbf{a} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ is

$$\frac{\mathbf{a}^\top (\mathbf{b} + \mathbf{c})}{|\mathbf{b} + \mathbf{c}|} = \frac{\lambda + 2 + 6 + 4}{\sqrt{(\lambda + 2)^2 + 40}} = \frac{\lambda + 12}{\sqrt{(\lambda + 2)^2 + 40}} = 1 \tag{1.2.20.2}$$

Squaring,

$$(\lambda + 12)^2 = (\lambda + 2)^2 + 40 \implies 20\lambda = -100 \implies \lambda = -5 \tag{1.2.20.3}$$

Check: $\lambda + 12 = 7 > 0$ and $\sqrt{9 + 40} = 7$, so the unsquared equation holds. The following code verifies the solution

codes/31A.c

1.2.21 Find the shortest distance between the lines $\mathbf{r} = \left(2\hat{i} - \hat{j} + 3\hat{k}\right) + \lambda\left(\hat{i} - 2\hat{j} + 3\hat{k}\right)$ and $\mathbf{r} = \left(\hat{i} + 4\hat{k}\right) + \mu\left(3\hat{i} - 6\hat{j} + 9\hat{k}\right)$. (CBSE Class 12, 65/7/1, Q.31(B))

Solution: The direction vectors $\mathbf{d}_1 = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} 3 \\ -6 \\ 9 \end{pmatrix} = 3\mathbf{d}_1$ are proportional, so the lines are parallel. Take $\mathbf{P}_1 = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ and $\mathbf{P}_2 = \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix}$, so $\mathbf{P}_2 - \mathbf{P}_1 = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$. The distance between parallel lines is

$$d = \frac{\|(\mathbf{P}_2 - \mathbf{P}_1) \times \mathbf{d}_1\|}{\|\mathbf{d}_1\|} \tag{1.2.21.1}$$

$$\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \\ 1 \end{pmatrix}, \quad d = \frac{\sqrt{25+16+1}}{\sqrt{1+4+9}} = \frac{\sqrt{42}}{\sqrt{14}} = \sqrt{3} \tag{1.2.21.2}$$

The following code verifies the solution

codes/31B.c

1.2.22 Evaluate $\int \frac{x^2+1}{(x^2+2)(2x^2+1)}\,dx$. (CBSE Class 12, 65/7/1, Q.32(A))

Solution: Partial fractions in x^2:

$$\frac{x^2+1}{(x^2+2)(2x^2+1)} = \frac{A}{x^2+2} + \frac{B}{2x^2+1} \implies x^2+1 = A\left(2x^2+1\right) + B\left(x^2+2\right) \tag{1.2.22.1}$$

Comparing coefficients, $2A + B = 1$ and $A + 2B = 1$, so $A = B = \frac{1}{3}$. Using $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a}$,

$$\int \frac{x^2+1}{(x^2+2)(2x^2+1)}\,dx = \frac{1}{3\sqrt{2}}\tan^{-1}\left(\frac{x}{\sqrt{2}}\right) + \frac{1}{3\sqrt{2}}\tan^{-1}\left(\sqrt{2}\,x\right) + C \tag{1.2.22.2}$$

The program compares a midpoint Riemann sum over $[0,1]$ with $F(1) - F(0) \approx 0.37024$. The following code verifies the solution

codes/32A.py

The following code generates Fig. 1.2.22.1.

codes/32Aplot.py

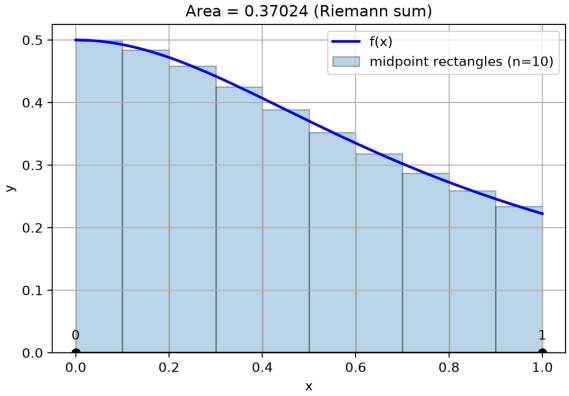


Fig. 1.2.22.1: Midpoint Riemann sum for $f\left(x\right)$ on $\left[0,1\right]$

1.2.23 Evaluate $\int_0^{\pi} \frac{x\tan x}{\sec x+\tan x}\,dx$. (CBSE Class 12, 65/7/1, Q.32(B))

Solution: Since $\frac{\tan x}{\sec x+\tan x} = \frac{\sin x}{1+\sin x}$, the integral is $I = \int_0^{\pi} \frac{x\sin x}{1+\sin x}\,dx$. Using $\int_0^{\pi} xg\left(x\right)\,dx = \frac{\pi}{2}\int_0^{\pi} g\left(x\right)\,dx$ when $g\left(\pi-x\right) = g\left(x\right)$,

$$I = \frac{\pi}{2}\int_0^{\pi} \frac{\sin x}{1+\sin x}\,dx = \frac{\pi}{2}\int_0^{\pi}\left(1-\frac{1}{1+\sin x}\right)dx \tag{1.2.23.1}$$

Rationalising, $\frac{1}{1+\sin x} = \sec^2 x - \sec x\tan x$, whose antiderivative is $\tan x - \sec x$. Over $\left[0,\pi\right]$ this gives $\int_0^{\pi} \frac{dx}{1+\sin x} = 2$ (the point $x = \frac{\pi}{2}$ is handled by taking limits on either side). Hence

$$I = \frac{\pi}{2}\left(\pi-2\right) \approx 1.7932 \tag{1.2.23.2}$$

The following code verifies the solution

codes/32B.py

The following code generates Fig. 1.2.23.1.

codes/32Bplot.py

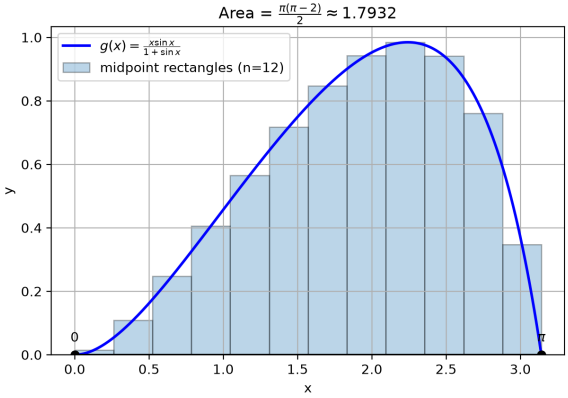


Fig. 1.2.23.1: Midpoint Riemann sum for $\frac{x \sin x}{1+\sin x}$ on $[0, \pi]$

1.2.24 Solve $\frac{dy}{dx} = \cos x - 2y$ using Laplace transforms. (CBSE Class 12, 65/7/1, Q.34)

Solution: Rewrite as $y' + 2y = \cos x$ and let $Y(s) = \mathcal{L}\{y\}$, $y(0) = y_0$. Since $\mathcal{L}\{y'\} = sY - y_0$ and $\mathcal{L}\{\cos x\} = \frac{s}{s^2+1}$,

$$(s+2)\,Y(s) = y_0 + \frac{s}{s^2+1} \implies Y(s) = \frac{y_0}{s+2} + \frac{s}{(s+2)(s^2+1)} \tag{1.2.24.1}$$

Partial fractions: $\frac{s}{(s+2)(s^2+1)} = \frac{-\frac{2}{5}}{s+2} + \frac{\frac{2}{5}s+\frac{1}{5}}{s^2+1}$. Using $\mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} = e^{-2x}$, $\mathcal{L}^{-1}\left\{\frac{s}{s^2+1}\right\} = \cos x$ and $\mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} = \sin x$,

$$y(x) = \left(y_0 - \frac{2}{5}\right)e^{-2x} + \frac{2}{5}\cos x + \frac{1}{5}\sin x = Ce^{-2x} + \frac{2\cos x + \sin x}{5} \tag{1.2.24.2}$$

No initial condition is given, so the constant C is arbitrary. The following code verifies the solution

codes/34.py

1.2.25 Find the points on the line $\frac{2x+4}{6} = \frac{y+1}{2} = \frac{-2z+6}{-4}$ at a distance of $3\sqrt{2}$ units from the point $P(1,2,3)$. (CBSE Class 12, 65/7/1, Q.35(A))

Solution: Let the common ratio be t. Then $x = 3t-2$, $y = 2t-1$, $z = 2t+3$. The squared distance from P is

$$(3t-3)^2 + (2t-3)^2 + (2t)^2 = 17t^2 - 30t + 18 = 18 \tag{1.2.25.1}$$

$$t(17t-30) = 0 \implies t = 0 \text{ or } t = \frac{30}{17} \tag{1.2.25.2}$$

The required points are

$$Q_1 = (-2, -1, 3), \qquad Q_2 = \left(\frac{56}{17}, \frac{43}{17}, \frac{111}{17}\right) \tag{1.2.25.3}$$

The following code verifies the solution

codes/35A.c

The following code generates Fig. 1.2.25.1.

codes/35Aplot.py

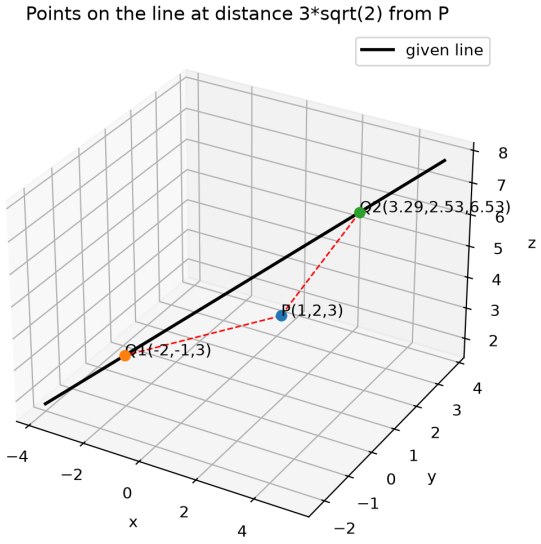


Fig. 1.2.25.1: Points Q_1, Q_2 on the line at distance $3\sqrt{2}$ from P

1.2.26 Find the image of the point $P\,(-1, 5, 2)$ in the line $\frac{2x-4}{2} = \frac{y}{2} = \frac{2-z}{3}$, and the length PP'. (CBSE Class 12, 65/7/1, Q.35(B))

Solution: Let the common ratio be t: $x = t + 2$, $y = 2t$, $z = 2 - 3t$. The line passes through $\mathbf{A} = \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$ with direction $\mathbf{d} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$. The foot of the perpendicular $H = \mathbf{A} + t_0\mathbf{d}$ has

$$t_0 = \frac{(\mathbf{P} - \mathbf{A})^\top \mathbf{d}}{\mathbf{d}^\top \mathbf{d}} = \frac{\begin{pmatrix} -3 \\ 5 \\ 0 \end{pmatrix}^\top \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}}{14} = \frac{7}{14} = \frac{1}{2} \implies \mathbf{H} = \begin{pmatrix} \frac{5}{2} \\ 1 \\ \frac{1}{2} \end{pmatrix} \tag{1.2.26.1}$$

Since H is the mid-point of PP',

$$\mathbf{P'} = 2\mathbf{H} - \mathbf{P} = \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix} - \begin{pmatrix} -1 \\ 5 \\ 2 \end{pmatrix} = \begin{pmatrix} 6 \\ -3 \\ -1 \end{pmatrix} \tag{1.2.26.2}$$

$$PP' = \sqrt{7^2 + 8^2 + 3^2} = \sqrt{122} \tag{1.2.26.3}$$

The following code verifies the solution

codes/35B.c

The following code generates Fig. 1.2.26.1.

codes/35Bplot.py

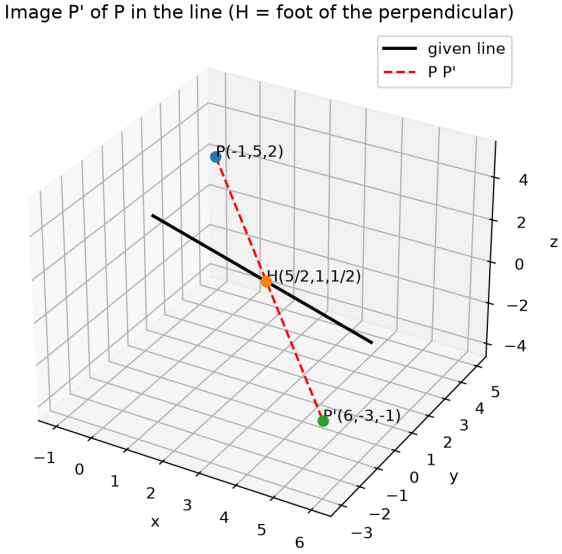


Fig. 1.2.26.1: Point P, its image P' and the foot H

1.2.27 **Case study (camphor).** The radius r of a camphor tablet decreases with time t (in hours) as it evaporates, and $\frac{dr}{dt} = -\frac{2}{3}k$ with $k > 0$ and $r\,(0) = 5$ mm. (i) Write the order and degree of the differential equation. (ii) Solve it using the Laplace transform. (iii)(a) If $r = 3$ mm after 1 hour, find the time for the radius to become 0. **OR** (iii)(b) If $r = 1$ mm after 1 hour, find the time for the radius to become 0. (CBSE Class 12, 65/7/1, Q.37)

Solution: (i) The highest derivative is $\frac{dr}{dt}$ and its power is 1, so the order is 1 and the degree is 1.

(ii) Let $R(s) = \mathcal{L}\{r\}$. Then

$$sR(s) - r(0) = -\frac{2k}{3s} \implies R(s) = \frac{5}{s} - \frac{2k}{3s^2} \implies r(t) = 5 - \frac{2}{3}kt \quad (1.2.27.1)$$

(iii)(a) From $r(1) = 3$: $3 = 5 - \frac{2}{3}k \implies k = 3$, so $r(t) = 5 - 2t$. Setting $r = 0$,

$$t = \frac{5}{2} \text{ hours} \quad (1.2.27.2)$$

(iii)(b) From $r(1) = 1$: $1 = 5 - \frac{2}{3}k \implies k = 6$, so $r(t) = 5 - 4t$ and

$$t = \frac{5}{4} \text{ hours} \quad (1.2.27.3)$$

The following code verifies the solution

codes/37.c

1.2.28 **Case study (health survey).** Of 1000 people, 700 have good health (A_1), 200 average health (A_2) and 100 poor health (A_3). The probabilities of contracting the disease D are $\Pr(D \mid A_1) = 0.25$, $\Pr(D \mid A_2) = 0.35$ and $\Pr(D \mid A_3) = 0.50$. (i) Find the probability that a randomly selected person has contracted the disease. (ii) Find the probability that a person who has not contracted the disease belongs to category A_1. (CBSE Class 12, 65/7/1, Q.38)

Solution: The mapping from category to probability is $\Pr(A_1) = 0.7$, $\Pr(A_2) = 0.2$, $\Pr(A_3) = 0.1$, and these sum to 1.

(i) By the law of total probability,

$$\Pr(D) = \sum_{i=1}^{3} \Pr(D \mid A_i) \Pr(A_i) = 0.25(0.7) + 0.35(0.2) + 0.50(0.1) = 0.295 \quad (1.2.28.1)$$

(ii) $\Pr(D') = 1 - 0.295 = 0.705$ and $\Pr(D' \mid A_1) = 0.75$. By Bayes' theorem,

$$\Pr(A_1 \mid D') = \frac{\Pr(D' \mid A_1) \Pr(A_1)}{\Pr(D')} = \frac{0.75(0.7)}{0.705} = \frac{0.525}{0.705} = \frac{35}{47} \approx 0.7447 \quad (1.2.28.2)$$

The following code verifies the solution

codes/38.c